HemanthKumar K Ramakrishnan

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SUMMARY

Senior Software Engineer with 8+ years of experience in designing distributed systems, event-driven microservices, and full-stack applications using Java, Python, TypeScript/React, Spring WebFlux, AWS, Kafka, Spark, and MongoDB. Proficient in building high-performance PySpark pipelines, automating data workflows with IICS, Bash, and Control-M, & deploying applications on AWS EKS. Experienced with Generative and Agentic AI using LangGraph, MCP, RAG, multi-agent orchestration, tool calling, and LLM-based automation to streamline engineering workflows.

SKILLS

Java, Python, PySpark, Spark, R | TypeScript, React | SQL, No-SQL | Spring Boot, Spring WebFlux | AWS | Kafka | Kubernetes, Docker | Redis, MongoDB | RAG, Agentic AI, LangGraph, CrewAI, MCP, A2A, Multi-Agent Systems, Langfuse | Informatica, Control-M | Splunk, Kibana

EXPERIENCE

Senior Software Engineer

Freddie Mac (Client) - Thinklusive Inc (FTE) **May 2020 - Present, McLean, Virginia**

- Led delivery of a web application built on Java 17, Spring WebFlux and TypeScript with React for processing 1M+ appraisals monthly on AWS EKS.
- Built Java backend to process S3 XML via SQS, enrich through external APIs, consume SLID updates via Kafka, and persist JSON to MongoDB.
- Designed React/TypeScript file uploader with Tailwind CSS for appraisal submissions, reducing upload errors by 75% and latency by 30%.
- Delivered GraphQL & gRPC APIs over MongoDB, secured with Apigee and OAuth2, enabling sub-second appraisal retrieval for 27+ clients.
- Built RAG incident-triage agent using Confluence pages & codebase to answer project queries and accelerate L3 production troubleshooting.
- Architected CrewAI based financial agents using enterprise APIs to coordinate research, valuation, and reporting, reducing manual analysis.
- Built LangGraph/MCP agent for converting business requirements into Jira stories with acceptance criteria & automated PRs after code commits.
- Implemented agent evaluation pipelines using Langfuse & OTel to benchmark task accuracy, tool-call latency, and reliability across workflows.
- Developed scalable PySpark jobs on AWS EMR to enrich 1M+ records per batch with scrubbed addresses and external metadata sources.
- Automated daily NAS-to-S3 file transfers using Control-M file watchers and Informatica CDI, completely removing manual processing steps.

Software Engineer

Labcorp (Client) - Thinklusive Inc (FTE) **June 2019 - May 2020, Durham, North Carolina**

- Developed a robust Java Spring Boot application to retrieve patient and service center data from a DB2 database for healthcare reporting.
- Implemented caching using Spring Cache and Redis to reduce database calls and improve response time for frequently accessed endpoints.
- Built Random Forest classifier using XGBoost feature importance to predict patient wait-time delays and support service center scheduling.
- Built scalable R Shiny web app on Cloudera for self-service dashboards, increasing stakeholder satisfaction by 40% with real-time insights.
- Migrated legacy data from Oracle database to Hive using Sqoop and HiveQL, significantly reducing batch runtimes from 45 to 12 minutes.
- Automated datameer workbook generation for genetics analytics, cutting report delivery from 3 days to under 2 hours for stakeholders.

Software Engineer

VentureSpace **August 2018 - May 2019, New York, New York**

- Built robust Python web scrapers aggregating listings from 10+ sites daily, tripling data coverage and reducing manual research time by 80%.
- Deployed scalable FastAPI microservice with Pydantic models and 95% test coverage, delivering live valuations under 50 ms for broker tools.
- Optimized complex PostgreSQL queries and materialized views, cutting dashboard load times from 12s to 1.5s for real-time trend analysis.
- Created 15+ interactive dashboards in Superset and Tableau, surfacing key business metrics that drove a 20% boost in listing conversions.

EDUCATION

Master of Science : Information Systems

New Jersey Institute of Technology • New Jersey, USA • 2019 • 3.65 GPA

Bachelors of Engineering : Electronics and Communication Engineering

Anna University • Chennai, India • 2017 • 3.2 GPA

CERTIFICATIONS

Claude Certified Architect (CCAR-F)

Anthropic • 2026

AI Engineer Agentic Track by Ed Donner

Udemy • 2026

AWS Certified Developer – Associate

AWS • 2024